Project Initialization and Planning Phase

Date	01 October 2026
Team ID	SWTID-2026-5883
Project Title	Pocket Smart AI: Your Smart Budget & Recommendation Assistant
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

PocketSmart AI is a GenAI-powered, cross-platform recommendation system designed to help users plan budgets and receive personalized product and service suggestions. The project uses budget, preferences, contextual information, and optional images to generate recommendations across home interior planning, party planning, and jewelry selection. The source project document describes integration with Gemini 1.5 Flash Pro and platforms such as Amazon, Flipkart, IKEA, Swiggy, Zomato, and OYO.

Project Overview

Objective	To make budget planning simpler by generating personalized, budget-aware recommendations for everyday lifestyle needs across multiple categories.
Scope	Home Interior Planner, Party Budget Planner, and Jewelry Budget Planner, including budget inputs, preferences, optional outfit images, recommendations, authentication, sessions, and recommendation history.
Problem Statement	Managing budgets across different needs can be overwhelming because users must compare many products, services, platforms, and price ranges. The project aims to reduce financial guesswork and simplify decision-making.
Impact	The system is intended to save time, support budget-aware decisions, and provide organized recommendations tailored to the user's context and preferences.
Proposed Solution	A modular AI recommendation platform using Gemini 1.5 Flash Pro with a web frontend and backend APIs. It processes text, numbers, and optional images to generate structured recommendations.
Key Features	Budget-based planning; personalized recommendations; Home, Party, and Jewelry modules; optional image analysis for jewelry; user registration/login; session handling; recommendation history; multi-platform sourcing.

Resource Requirements

Resource Type	Description	Specification / Allocation
Hardware	**Development device and internet connectivity**	Laptop / desktop with reliable internet; exact CPU/GPU/RAM specification is not specified in the source project document.
Software	**Programming language and backend frameworks**	Python; FastAPI is used in the implementation milestones; Flask is also described in the project architecture.
AI Platform	**Generative AI model for budget interpretation, recommendations and image analysis**	Gemini 1.5 Flash Pro with API access and configured API key.
Frontend	**User interface for budgets, preferences, images and AI results**	HTML, CSS, JavaScript, Jinja2 templates.
Libraries / Server Tools	**Application support and server execution**	Python libraries required by the implementation; Uvicorn is used to start the FastAPI application.
External Platforms / APIs	**Product and service data sourcing**	Amazon, Flipkart, IKEA, Swiggy, Zomato, OYO and similar platforms; mock API calls or simulated scraping are also described for development.
Data / Inputs	**Information required for recommendations**	User budget, preferences, category details, guest count/event details, room details, occasion, style preferences, and optional outfit image.

Major Development Activities

1. Gemini AI Setup	Configure Gemini API access, keys, usage limits, multimodal testing, and prompts for budget interpretation and recommendations.
2. Core Functionality	Develop planner modules and endpoints for Home, Party, and Jewelry recommendations, with optional image input.
3. Backend Integration	Implement API routes, authentication, sessions, CORS, recommendation details, history, and startup configuration.
4. UI Development	Build responsive forms and structured result displays for the three planner modules.
5. Testing & Optimization	Test varied real-world budgets, evaluate recommendation quality and budget adherence, refine prompts, validations, and fallback recommendations.

Team Members and Work Allocation

S.No.	Name	Assigned Work	Main Responsibilities
1	Jashvini.K	**AI & Recommendation Engine**	Gemini 1.5 Flash Pro integration; prompt design for budget interpretation; recommendation logic for Home, Party and Jewelry planners; multimodal/image-input support.
2	Indhumathi.D	**Backend & API Development**	FastAPI backend routes; planner endpoints; authentication and session handling; CORS; recommendation and history APIs.
3	Jayapriya Rajkumar	**Frontend & UI Development**	HTML/CSS/JavaScript and Jinja2 interfaces; budget forms; category selection; result cards; responsive user interaction.
4	Jonal.A	**Planner Modules & Data Integration**	Home, Party and Jewelry planner input flows; third-party platform/product-service sourcing; mock API or simulated data integration; category-specific logic.
5	Kavibharathi.K	**Testing, Optimization & Documentation**	Real-world budget testing; recommendation-quality checks; validation and fallback testing; UI/UX verification; project documentation and final integration support.

Project Team

1. Jashvini.K	2. Indhumathi.D
3. Jayapriya Rajkumar	4. Jonal.A
5. Kavibharathi.K	Team ID: SWTID-2026-5883

Note: The work allocation above is assigned in the exact order of names provided by the team and is organized around the development activities documented in the supplied PocketSmart AI project material.